

Homework #7

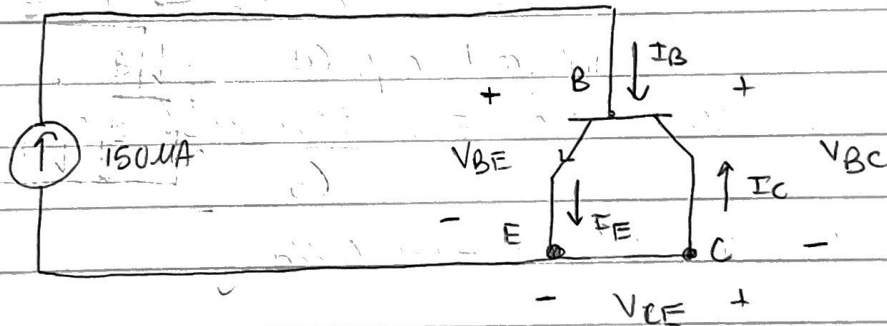
1. Perret 12.3 : DC operating point $\rightarrow (I_B = 5 \mu A, V_{EC} = 10 V)$

↳ Determine values for g_m and r_π @ DC operating point, $I_C \approx 1.1$

$$g_m = \frac{q I_C}{K T} = \frac{(1.1 \text{ mA})}{0.0259} \approx 4.25 \times 10^{-2} \text{ S (Siemens)} \rightarrow A/V$$

$$r_\pi = \frac{K T}{q I_B} = \frac{0.0259 \text{ V}}{5 \mu A} \approx 5.18 \times 10^3 \Omega$$

2. BJT : $I_S = 2 \times 10^{-15} \text{ A}$, $\beta_F = 100$, $\beta_R = 0.25$



b) npn transistor b/c arrow points away from base on emitter

d) $V_{BE} = V_{BC}$ b/c $V_C = V_E$ b/c of shared node

e) Transport Model Equations : b/c $V_{BE} = V_{BC} \therefore I_T = 0$

$$I_E = I_S \left[\underbrace{e^{q V_{BE} / K T} - e^{q V_{BC} / K T}}_{I_T = 0} \right] + \frac{I_S}{\beta_F} [e^{q V_{BE} / K T} - 1]$$

$$I_C = \frac{I_T}{\beta_F} - \frac{I_S}{\beta_F} [e^{q V_{BC} / K T} - 1]$$

$$I_B = I_S \left(\frac{1}{\beta_F} + \frac{1}{\beta_R} \right) [e^{q V_{BE} / K T} - 1] \rightarrow \text{cont.}$$

$$\frac{I_E}{I_B} = \frac{\left(\frac{I_S}{\beta_F} [e^{qV_{BE}/kT} - 1] \right)}{\left(\frac{I_S}{\beta_F} \left(\frac{1}{\beta_F} + \frac{1}{\beta_R} \right) [e^{qV_{BE}/kT} - 1] \right)} = \frac{\frac{1}{\beta_F}}{\frac{1}{\beta_F} + \frac{1}{\beta_R}} = \frac{1}{1 + \frac{\beta_F}{\beta_R}}$$

$$\frac{I_E}{I_C} = \frac{\left(\frac{I_S}{\beta_F} [e^{qV_{BE}/kT} - 1] \right)}{\left(-\frac{I_S}{\beta_R} [e^{qV_{BE}/kT} - 1] \right)} = \frac{-\beta_R}{\beta_F}$$

$$f) I_C = \left(\frac{-\beta_F}{\beta_R} \right) I_E \rightarrow I_B = I_E - I_C = I_E - \left(-\frac{100}{0.25} \right) I_E$$

$$\therefore I_B = 401 I_E \rightarrow I_E = \frac{I_B}{401}$$

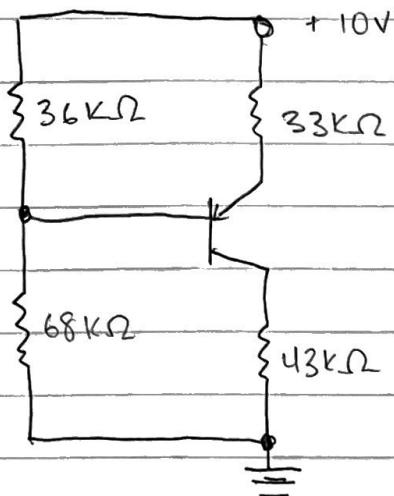
$$\text{where } I_B = 150 \mu A \rightarrow I_E = \frac{150 \mu A}{401} = 0.374 \mu A$$

$$I_C = -400 \times 0.374 \mu A = -149.6 \mu A$$

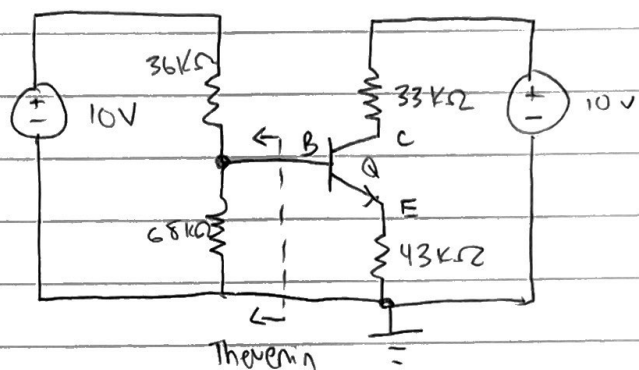
$$\text{Using } \frac{I_E}{\beta_F} = \frac{I_S}{\beta_F} [e^{V_{BE}/V_T} - 1] \rightarrow V_{BE} = \ln \left(\frac{I_E \beta_F}{I_S} \right) V_T$$

$$V_{BE} = V_{BC} = \ln \left(\frac{0.374 \mu A \cdot 100}{2 \times 10^{-15} A} \right) (0.025) \approx 0.591 V$$

3. Q-point for 4-resistor bias ckt : $\beta_F = 50$, $V_{BE} = 0.7V$

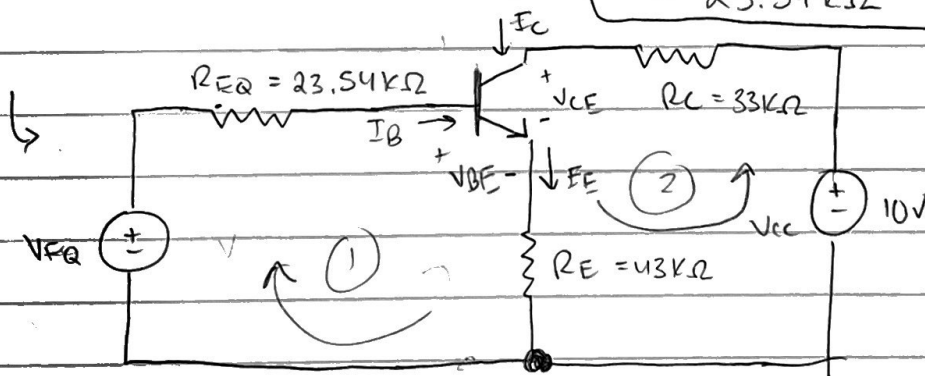


Split into 2 equal voltage sources



$$V_{EQ} = \frac{R_1}{R_1 + R_2} V_{CC} = \frac{68k\Omega}{36k\Omega + 68k\Omega} (10V) = 6.54V$$

$$R_{EQ} = \frac{V_{EQ}}{(V_{CC}/R_2)} = \frac{R_1 \times R_2}{R_1 + R_2} = \frac{36k\Omega \times 68k\Omega}{104k\Omega} = 23.54k\Omega$$



Loop 1: $0 = -V_{EQ} + R_{EQ} I_B + V_{BE} + R_E I_E$
 $\therefore V_{EQ} = R_{EQ} I_B + V_{BE} + R_E I_E$ } $V_{BE} = 0.7V$

$I_E = (\beta_F + 1) I_B$
 $I_C = \beta_F I_B$ } $I_B = 1.618 \mu A$
 $I_C = 50 I_B = 80.9 \mu A$
 $I_E = 51 I_B = 82.5 \mu A$

Loop 2: $V_{CC} = R_C I_C + V_{CE} + R_E I_E$ } $V_{CE} = 3.797V$

$I_C = \beta_F I_B$

$V_{BC} = -(V_{CE} - V_{BE}) =$

$\therefore Q\text{-point} \therefore (80.9 \mu A, 3.8V)$